

# RUTIKA AVINASH KADAM

Stony Brook, NY 11790 · +1 934-949-8653 · rutikakadam2727@gmail.com · LinkedIn · Portfolio · GitHub

## EDUCATION

### Stony Brook University

Master of Science in Data Science

**Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Large Language Models, Agentic AI, Big Data.

### Savitribai Phule Pune University

Bachelor of Engineering in Information Technology

**Coursework:** Discrete Structures, Data Structures & Algorithms, System Design, Cloud Computing.

Stony Brook, USA

August 2024 – May 2026

Pune, India

August 2016 – May 2020

## SKILLS

**Programming Languages:** Python, R, SAS, SQL (MySQL, PostgreSQL, PL/SQL), C++, HTML, CSS

**Data Science:** NumPy, Pandas, Seaborn, scikit-learn, TensorFlow, Keras, PyTorch, Hugging Face, LangChain, LangGraph

**Data Engineering & Big Data:** Azure Data Factory, Data Lake, Databricks, Microsoft Fabric, SSMS, SSIS

**Cloud & DevOps:** Azure, AWS, FastAPI, REST APIs, Docker, Kubernetes, MLflow, Git, GitHub, Postman

**Visualization & Analytics:** Power BI, Tableau, Microsoft Excel

**Security & IT Operations:** Qualys VMDR, Microsoft Endpoint Configuration Manager (SCCM), ServiceNow, Jira, Aternity,

Vulnerability Management, Patch Management

## PROFESSIONAL EXPERIENCE

### Stony Brook Medicine

Data Scientist (Graduate Researcher)

Stony Brook, USA

July 2025 – Present

- Built an end-to-end **Multi-Modal RAG pipeline** over **6k+ pages of medical documents** for Department of Public Health, Stony Brook Medicine, using **Azure Document Intelligence** for layout-aware extraction & **GPT-4o** for figure captioning.
- Implemented **custom semantic chunking** using embedding-based sentence similarity & adaptive breakpoint detection.
- Designed & benchmarked **FLAT, HNSW, & IVF\_FLAT indexes in Milvus** across **27k+ multi-modal chunks**, comparing p50/p95 latency & Recall@5; achieved comparable accuracy with approximate indexes at **~2.5× lower latency** than FLAT.
- Evaluated **vector search, MMR reranking, & hybrid Vector+BM25 retrieval with RRF** on a synthetic ground-truth dataset spanning text, tables, & images; achieved **0.90 Recall@5** with hybrid retrieval.
- Built a **grounded RAG generation pipeline** with structured JSON outputs & source citations, achieving **RAGAS scores of 0.82 faithfulness, 0.82 answer relevancy, 0.93 context precision, & 0.74 context recall**.

### Tata Consultancy Services

System Analyst

Pune, India

August 2020 – April 2024

- Collaborated with Vulnerability Management, performed **exploratory data analysis** on **1 Million+ system vulnerability records** reporting from **Qualys VMDR** using **Python & MySQL**, uncovering trends, anomalies, & threat vectors.
- Reduced **security risk exposure by 15%** by developing **Linear & Logistic Regression models** predicting exploitability & patch prioritization using engineered vulnerability features.
- Discovered recurring remediation patterns across **1M+ records** by applying **semantic clustering using E5 embeddings with HDBSCAN** on remediation text data.
- Fine-tuned **BERT classifier** on cluster-derived labels to predict remediation categories in real time, enabling automated ticket routing, cutting **manual triage by 40%**.
- Improved vulnerability monitoring visibility by designing **Power BI dashboards with DAX-driven KPIs** including **CVSS score, exploitability risk, remediation timelines, & compliance metrics**.
- Achieved 99% patch compliance across enterprise systems by deploying automated remediation workflows through Microsoft Endpoint Configuration Manager & automated tasks using PowerShell, boosting productivity by 25%.

### Zensar Technologies

Data Intern

Pune, India

May 2019 – July 2019

- Implemented **ETL pipelines in Azure Data Factory with incremental batch loading** to ingest **multi-channel website traffic & operational data** into **Azure Data Lake Gen2** for scalable data integration for **Swiggy-India**.
- Developed **Power BI dashboards** to visualize **conversion funnel metrics & marketing channel attribution**, identifying drivers of traffic variability & campaign performance, increasing **high-intent traffic by 15%**.

## PROJECT EXPERIENCE

**Cardiovascular Disease Prediction** | Deep Learning, TensorFlow, Keras, Scikit-learn, Streamlit, Docker

- Achieved **81% model accuracy (AUC 0.87)** by developing a **hybrid deep learning architecture combining Artificial Neural Network & Convolution Neural Network** for cardiovascular disease prediction.
- Improved **early cardiac-risk detection with recall of 0.94** & deployed an **interactive Streamlit application on Hugging Face Spaces** for real-time predictions.

**Airbnb Price Prediction** | Pandas, Numpy, Seaborn, Matplotlib, Scikit-learn, Machine Learning, Bagging & Boosting

- Identified **key pricing drivers across Seattle Airbnb listings** through **exploratory data analysis** of location, room type, amenities, & occupancy factors.
- Improved price prediction performance by benchmarking multiple regression models and achieving best performance with **Random Forest ( $R^2 = 0.68$ )**.